

TryHackMe Zeek-Based Investigation: Phishing Alert

1. Introduction

In this case, a Zeek alert flagged a 'Phishing Attempt'. As a SOC analyst, the objective is to analyze the associated PCAP file using Zeek to validate the alert, identify Indicators of Compromise (IOCs), and gather evidence for reporting.

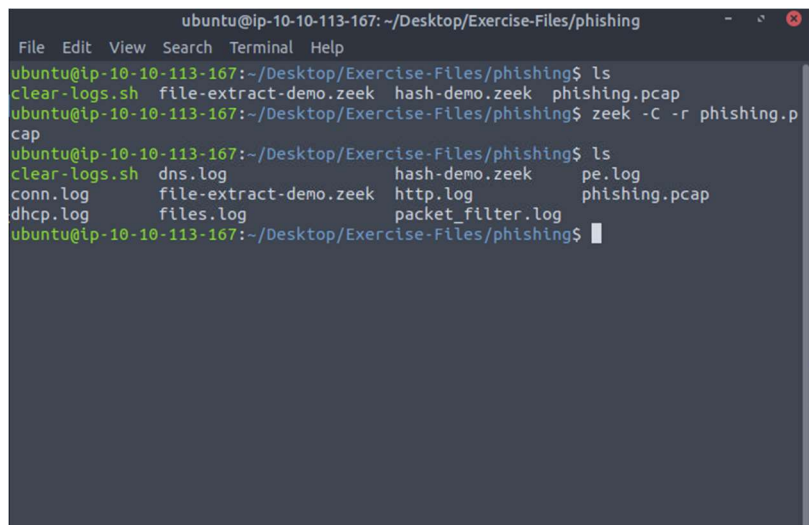
2. Objective

Confirm the legitimacy of the phishing alert and collect relevant IOCs such as source IP addresses, domains, and file hashes associated with the malicious activity.

3. Investigation Steps

Navigating to the PCAP Directory

First I navigated to the PCAP Directory `cd Desktop/Exercise-Files/Phishing`, then I used Zeek on the PCAP file using `zeek -C -r phishing.pcap` and then I used 'ls' to list the generated files as shown below:



```
ubuntu@ip-10-10-113-167: ~/Desktop/Exercise-Files/phishing
File Edit View Search Terminal Help
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ ls
clear-logs.sh file-extract-demo.zeek hash-demo.zeek phishing.pcap
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ zeek -C -r phishing.pcap
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ ls
clear-logs.sh dns.log hash-demo.zeek pe.log
conn.log file-extract-demo.zeek http.log phishing.pcap
dhcp.log files.log packet_filter.log
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$
```

Inspecting Connection Log (conn.log) for Unique Source IPs

```
cat conn.log | zeek-cut id.orig_h | sort | uniq
```

```

ubuntu@ip-10-10-113-167: ~/Desktop/Exercise-Files/phishing
File Edit View Search Terminal Help
conn.log file-extract-demo.zeek http.log phishing.pcap
dhcp.log files.log packet_filter.log
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ cat conn.log | head -
n 10
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path conn
#open 2025-07-30-08-48-32
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p
p proto service duration orig_bytes resp_bytes conn_sta
te local_orig local_resp missed_bytes history orig_pkts o
rig_ip_bytes resp_pkts resp_ip_bytes tunnel_parents
#types time string addr port addr port enum string interval
count count string bool bool count string count count c
ount set[string]
1561667867.459349 CpkUd23C7aJ07j6j5 10.6.27.102 50800 224.0.0.
252 5355 udp dns 0.109137 44 0 S0 -
0 D 2 100 0 0 -
1561667868.993491 C0lCnE4fluxMGL1ZZd 10.6.27.102 56900 10.6.27.
1 53 udp dns 0.025235 34 50 SF -
0 Dd 1 62 1 78 -
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ cat conn.log | zeek-c
ut id.orig_h | sort | uniq
10.6.27.102
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$

```

Defanging Suspicious IP (Using CyberChef)

Original IP: 10.6.27.102 → Defanged: 10[.]6[.]27[.]102

Examining HTTP Log for Malicious Domains

cat http.log | zeek-cut id.orig_h host

```

ubuntu@ip-10-10-113-167: ~/Desktop/Exercise-Files/phishing
File Edit View Search Terminal Help
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ cat http.log | head -
n 10
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path http
#open 2025-07-30-08-48-32
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p
p trans_depth method host uri referrer version user_age
nt origin request_body_len response_body_len status_code s
tatus_msg info_code info_msg tags username password
proxied orig_fuids orig_filenames orig_mime_types resp_fuids
enames resp_mime_types
#types time string addr port addr port count string string s
tring string string string count count count string count s
tring set[enum] string string set[string] vector[string] vector[s
tring] vector[string] vector[string] vector[string]
1561667874.713411 CC0c6KeQYLbRU0GVl 10.6.27.102 49157 23.63.25
4.163 80 1 GET www.msftncsi.com /ncsi.txt - 1
.1 Microsoft NCSI - 0 14 200 OK - (
empty) - - - Fpgan59p6uvNzLFja -
text/plain
1561667889.643717 CifzDi2AcTe07nnT37 10.6.27.102 49159 107.180.
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$ cat http.log | zeek-cut host id.or
ig_h
www.msftncsi.com 10.6.27.102
smart-fax.com 10.6.27.102
smart-fax.com 10.6.27.102
ubuntu@ip-10-10-113-167:~/Desktop/Exercise-Files/phishing$

```

Defanging Suspicious Domain (Using CyberChef)

Original Domain: smart-fax.com → Defanged: smart-fax[.]com

Extracting File Hashes Using Zeek Script

zeek -C -r phishing.pcap hash-demo.zeek

```

ubuntu@ip-10-10-157-58: ~/Desktop/Exercise-Files/phishing
File Edit View Search Terminal Help
ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$ zeek -C -r phishing.pcap
ap
ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$ ls
clear-logs.sh  dns.log      hash-demo.zeek  pe.log
conn.log      file-extract-demo.zeek  http.log      phishing.pcap
dhcp.log      files.log    packet_filter.log
ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$ cat hash-demo.zeek
# Enable MD5, SHA1 and SHA256 hashing for all files.

@load /opt/zeek/share/zeek/policy/frameworks/files/hash-all-files.zeek
ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$ cat /opt/zeek/share/zeek/policy/frameworks/files/hash-all-files.zeek
##! Perform MD5 and SHA1 hashing on all files.

@load base/files/hash

event file_new(f: fa_file)
{
  Files::add_analyzer(f, Files::ANALYZER_MD5);
  Files::add_analyzer(f, Files::ANALYZER_SHA1);
}
ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$

```

```

ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$ cat files.log
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path files
#open 2025-07-30-13-39-11
#fields ts      fuid      tx_hosts      rx_hosts      conn_uids      source d
ephth analyzers mime_type      filename      duration      local_or
ig is_orig seen_bytes total_bytes missing_bytes overflow_bytes t
imedout parent_fuid md5 sha1 sha256 extracted      extracted_cutoff
extracted_size
#types time string cat[add-1 cat[add-1 cat[add-1 string s

```

Filtering Hashes from file.log

cat files.log | zeek-cut mime_type filename md5 sha1

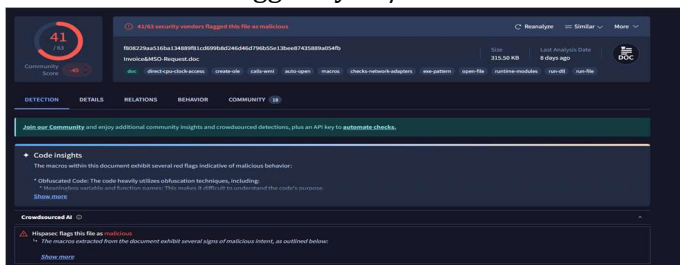
```

ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$ cat files.log | zeek-cut mime_type filename md5 sha1
text/plain - cd5a4d3fdd5bffc16bf959ef75cf37bc 33bf88d5b82df372
3d5863c7d23445e345828904
application/msword - b5243ec1df7d1d5304189e7db2744128 a66bd255
7016377dfb95a87c21180e52b23d2e4e
application/x-dosexec - cc28e40b46237ab6d5282199ef78c464 0d5c8200
02cf93384016bd4a2628dcc5101211f4
ubuntu@ip-10-10-157-58:~/Desktop/Exercise-Files/phishing$

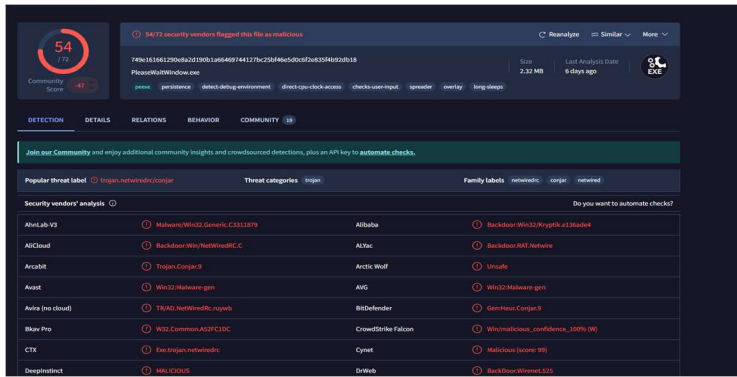
```

Analyzing MD5 on VirusTotal

Document MD5: Flagged by 41/63 vendors



Executable MD5: PleaseWaitWindow.exe - Flagged by 54/72 vendors



4. Indicators of Compromise (IOCs)

Type	Value
IP Address	10[.]6[.]27[.]102
Domain	smart-fax[.]com
Malicious Document MD5	b5243ec1df7d1d5304189e7db2744128
Executable MD5	cc28e40b46237ab6d5282199ef78c464

5. Recommendations

- Block the IP address 10[.]6[.]27[.]102 at the firewall.
- Add smart-fax[.]com to the domain blocklist.
- Perform a threat hunt across your environment for file hashes.
- Educate users on phishing risks and safe email practices.
- Implement endpoint detection and response (EDR) for future detection.

6. Conclusion

This investigation confirms the phishing alert was a true positive. With Zeek, we were able to identify and extract IOCs that can be used for further threat mitigation and response.